(3) $f'(x)=\dfrac{x^2-2x+3}{(x-1)^2}$ and hence the critical point of the curve occurs at $x=1$ as $f'(1)$ does not exist. But $x^2-2x+3=0$ has no real solution. Hence the only critical point occurs at $x=1$.

(4) $x=1$ is not in the domain of the function and $f'(x)\neq 0\ \forall\ x\in\mathbb{R}\setminus\{1\}$, there is no local maximum or local minimum.

(5) $f''(x)=-\dfrac{4}{(x-1)^3}\ \forall x\in\mathbb{R}\setminus\{1\}$. Therefore when $x<1, f''(x)>0$ the curve is concave upwards in $(-\infty,1)$ and when $x>1, f''(x)<0$ the curve is concave downwards in $(1,\infty)$. Since $x=1$ is not in the domain $f''(x)\neq 0\ x\in\mathbb{R}\setminus\{1\}$ there is no point of infection for $f(x)$.

(6) Since, $\lim\limits_{x\to 1^-}\dfrac{x^2-3x}{(x-1)}=+\infty$ and $\lim\limits_{x\to 1^+}\dfrac{x^2-3x}{(x-1)}=-\infty, x=1$ is a vertical asymptote.

The rough sketch is shown on the right side. ■

Example 7.72

Sketch the graph of the function $y=\dfrac{3x}{x^2-1}$.

Solution

(1) The domain of $f(x)$ is $\mathbb{R}\setminus\{-1,1\}$.

(2) Since $f(-x,-y)=f(x,y)$, the curve is symmetric about the origin.

(3) Putting $y=0$, we get $x=0$. Hence the x-intercept is $(0,0)$.

(4) Putting $x=0$, we get $y=0$. Hence the y-intercept is $(0,0)$.

(5) To determine monotonicity, we find the first derivative as $f'(x)=\dfrac{-3(x^2+1)}{(x^2-1)^2}$.

Hence, $f'(x)$ does not exist at $x=-1,1$. Therefore, critical numbers are $x=-1,1$. The intervals of monotonicity is tabulated in Table 7.9.

Interval	(-∞, -1)	(-1, 1)	(1, ∞)
Sign of $f'(x)$	–	–	–
Monotonicity	strictly decreasing	strictly decreasing	strictly decreasing

Table 7.9

(6) Since there is no sign change in $f'(x)$ when passing through critical numbers. There is no local extrema.

(7) To determine the concavity, we find the second derivative as $f''(x)=\dfrac{6x(x^2+3)}{(x^2-1)^3}$.

$f''(x)=0\Rightarrow x=0$ and $f''(x)$ does not exist at $x=-1,1$.

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